

Taller 2 - Ambientes virtuales

Machine Learning Engineering
Carlos Valladares - 221164

1. Repositorio

<https://github.com/vgcarlol/Machine-Learning-Engineering/tree/main/taller2>

En la carpeta **taller2** están el **requirements.txt** con paquetes de ML, el pipeline de la actividad 1 con su propio **requirements.txt** y el **environment.yml** para Conda.

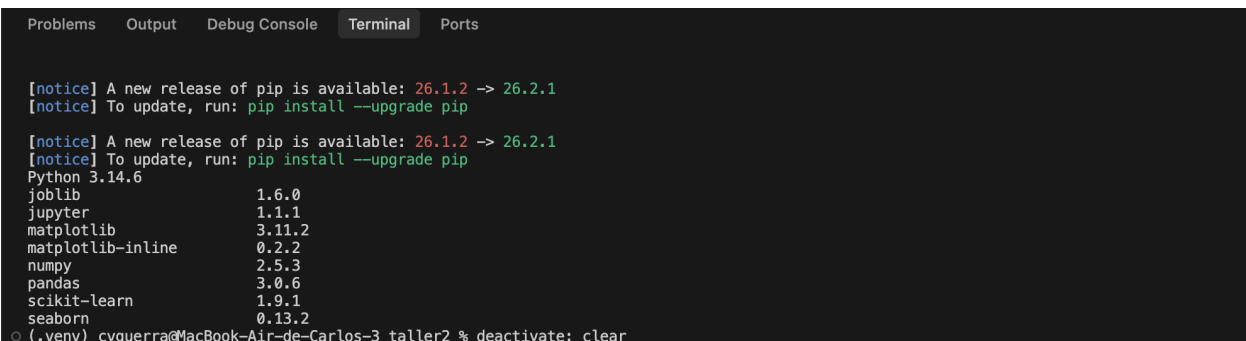
Para el reto del punto 3, el **pyproject.toml** del pipeline lee las dependencias desde **requirements.txt** y se agregó un **MANIFEST.in** para que el archivo se incluya al empaquetar:

```
[project]
dynamic = ["dependencies"]

[tool.setuptools.dynamic]
dependencies = { file = ["requirements.txt"] }
```

2. Ambiente de Python (venv)

```
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```



```
Problems  Output  Debug Console  Terminal  Ports

[notice] A new release of pip is available: 26.1.2 -> 26.2.1
[notice] To update, run: pip install --upgrade pip

[notice] A new release of pip is available: 26.1.2 -> 26.2.1
[notice] To update, run: pip install --upgrade pip
Python 3.14.6
joblib                1.6.0
jupyter               1.1.1
matplotlib            3.11.2
matplotlib-inline     0.2.2
numpy                 2.5.3
pandas                3.0.6
scikit-learn          1.9.1
seaborn               0.13.2
○ (.venv) cvguerra@MacBook-Air-de-Carlos-3 taller2 % deactivate; clear
```

3. Ambiente con Conda

```
conda env create -f environment.yml
conda activate taller2-ml
```

```
○ (.venv) cvguerra@MacBook-Air-de-Carlos-3 taller2 % deactivate; clear
eval "$(/opt/homebrew/Caskroom/miniconda/base/bin/conda shell.zsh hook)"
export CONDA_DEFAULT_CHANNELS=https://conda.anaconda.org/conda-forge
conda env create -f environment.yml
conda activate taller2-ml && python --version && conda env list
Retrieving notices: done
Channels:
- defaults
- nodefaults
Platform: osx-arm64
Collecting package metadata (repodata.json): done
Solving environment: done

## Package Plan ##

environment location: /opt/homebrew/Caskroom/miniconda/base/envs/taller2-ml

added / updated specs:
- joblib
- jupyter
- matplotlib
- numpy
- pandas
- pip
- python=3.12
- scikit-learn
- seaborn

The following packages will be downloaded:
```

package	build	
_openmp_mutex-4.5	8_kmp_llvm	8 KB
_python_abi3_support-1.0	hd8ed1ab_3	8 KB
anyio-4.15.1	pyh5ded981_1	171 KB
appnope-1.0.0	pyhcf101f3_0	12 KB

```
Machine Learning Engineering

Problems Output Debug Console Terminal Ports

(.venv) cvguerra@MacBook-Air-de-Carlos-3 taller2 % deactivate; clear
eval "$(/opt/homebrew/Caskroom/miniconda/base/bin/conda shell.zsh hook)"
export CONDA_DEFAULT_CHANNELS=https://conda.anaconda.org/conda-forge
conda env create -f environment.yml
conda activate taller2-ml && python --version && conda env list
from scikit-learn>=1.0->actividad1=0.1.0->-r /Users/cvguerra/Documents/GitHub/Machine Learning Engineering/taller2/condaenv.lynifnqj.requirements.txt (line 1)) (3.7.0)
Requirement already satisfied: cloudpickle>=3.0 in /opt/homebrew/Caskroom/miniconda/base/envs/taller2-ml/lib/python3.12/site-packages (from joblib>=1.4.0->scikit-learn>=1.0->actividad1=0.1.0->-r /Users/cvguerra/Documents/GitHub/Machine Learning Engineering/taller2/condaenv.lynifnqj.requirements.txt (line 1)) (3.1.2)
Requirement already satisfied: python-dateutil>=2.8.2 in /opt/homebrew/Caskroom/miniconda/base/envs/taller2-ml/lib/python3.12/site-packages (from pandas->actividad1=0.1.0->-r /Users/cvguerra/Documents/GitHub/Machine Learning Engineering/taller2/condaenv.lynifnqj.requirements.txt (line 1)) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in /opt/homebrew/Caskroom/miniconda/base/envs/taller2-ml/lib/python3.12/site-packages (from python-dateutil>=2.8.2->pandas->actividad1=0.1.0->-r /Users/cvguerra/Documents/GitHub/Machine Learning Engineering/taller2/condaenv.lynifnqj.requirements.txt (line 1)) (1.17.0)
Building wheels for collected packages: actividad1
  Building editable for actividad1 (pyproject.toml): started
  Building editable for actividad1 (pyproject.toml): finished with status 'done'
  Created wheel for actividad1: filename=actividad1-0.1.0-0.editable-py3-none-any.whl size=1339 sha256=956a344e18d97bba92588670618b5f4738ac314093ec3d44dbf03519a472edc5
  Stored in directory: /private/var/folders/vc/_1vltbs6gnbplfzc4ftytd80000gn/T/pip-ephem-wheel-cache-dv0p4nrx/wheels/19/e1/ba/ef8db0ab415d15dd2dc98d024a4f2f79c64005d3544bd4d932
Successfully built actividad1
Installing collected packages: actividad1
Successfully installed actividad1-0.1.0

done
#
# To activate this environment, use
#
# $ conda activate taller2-ml
#
# To deactivate an active environment, use
#
# $ conda deactivate

WARNING conda.pypi.main:notify_externally_managed_future(156):
Did you know? You can install many PyPI packages with conda
using the conda-pypi beta. Get started:
https://docs.conda.io/projects/conda/en/stable/new-features.html

Python 3.12.14

# conda environments:
#
# * -> active
base                /opt/homebrew/Caskroom/miniconda/base
taller2-ml          * /opt/homebrew/Caskroom/miniconda/base/envs/taller2-ml

○ (taller2-ml) cvguerra@MacBook-Air-de-Carlos-3 taller2 %
```

4. Conclusiones

Los ambientes virtuales permiten que cada proyecto tenga sus propias versiones de paquetes. En Machine Learning esto importa porque un cambio de versión en scikit-learn o pandas puede cambiar resultados o romper un modelo guardado.

El requirements.txt con versiones fijas hace que el entrenamiento se pueda repetir en otra computadora o en un servidor y obtener el mismo ambiente con un solo comando.

Conda es útil cuando el proyecto necesita dependencias que no son de Python, como librerías para GPU, y su environment.yml guarda también la versión de Python.

Incluir el archivo de requisitos dentro del paquete del pipeline evita que las dependencias queden repetidas en dos lugares y facilita pasar el pipeline a producción o a un contenedor.